

Supplementary Materials for  
**Threefold reduction of modeled uncertainty in direct radiative effects over  
biomass burning regions by constraining absorbing aerosols**

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Supplementary Text S1 and S2  
Figs. S1 to S25  
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## Supplementary Text

### Supplementary Text 1. SSA errors and impacts of selected factors

As the modeled MAC is closely related to SSA, we investigate here the SSA errors in the AeroCom models. In **Fig. S6**, the modeled relationship between SSA and rBC is compared with in situ observations by ref (24), with most models showing underestimated SSA for a given rBC. The underestimation of SSA in the models becomes even more significant with increasing rBC. The highly absorbing aerosols in Africa therefore suffer more from such an error than those in the Amazon. This error is unlikely to be addressed by adjusting the rBC in emissions, as done by previous studies (62).

Essentially, the relationship between SSA and rBC can be affected by several factors. In this study, we aim to investigate the impacts of three key factors, namely particle size distribution, complex refractive index for BC, and mixing state, on this relationship. To accomplish this, we employ a series of Mie calculations using the Mätzler (63) model to determine the cross section of aerosol extinction and absorption for mixtures of BC and OA with varying combinations of the aforementioned factors. We assume a spherical particle shape for all Mie calculations, as suggested by the low depolarization ratios from the Cloud Aerosol Lidar with Orthogonal Polarization (CALIOP) observations (64). The hygroscopic growth is considered using the Köhler theory (65), which is found to have little impact on the calculations. For mixing states, we consider three assumptions used in AeroCom models (see **Table S1**): 1) external mixing, which assumes BC and OA particles exist separately; 2) homogeneous internal mixing, which assumes BC and OA particles are mixed at a molecular level, and the volume-weighted averages are used to calculate the refractive index of the mixed particles; and 3) core-shell structure with BC as the core in the center and other components (e.g., OA) as the coated shell of the particles. In general, the latter two cases are both referred to as internal mixing.

As shown in **Fig. S2A2**, particle size distribution is a crucial determinant of the relationship between SSA and rBC, where smaller particles tend to exhibit greater absorption. This partly accounts for the SSA underestimation in most models, as they tend to produce particles that are too small (31). In addition, the BC refractive index can directly affect SSA (**Fig. S2B2**). These two factors combined can generally explain the opposite SSA errors in the ECHAM-HAM and SPRINTARS models, with the former producing small particle sizes (high AE) and high refractive index, and the latter featuring large sizes (the lowest AE in the AeroCom ensemble) and the lowest refractive index. Regarding the impacts of mixing state, both the homogeneous internal mixing and core-shell structure produce lower SSAs than external mixing. Such a difference stands out particularly with a larger particle and higher refractive index. Although the external mixing shows better agreement (higher SSA) with the observations, in situ and laboratory measurements have frequently confirmed the internal-mixing structure for BBA (66-70). In addition, models assuming an external mixing do not show superior performance compared with other models. For the two internal mixing states, small differences are found in the predicted SSA with a low BC fraction, as reported by a previous study (71). Within the range of refractive indices used by the AeroCom models (**Table S1**), both internal mixing states still fail to reproduce the observed SSA. We further reduce the imaginary part of the BC refractive index to 0.3, as suggested by field measurements (58,59), which reduces the SSA error. The impacts of different refractive indices are also tested in ECHAM-HAM via sensitivity tests (**Fig. S23**), where the best agreement between the model and

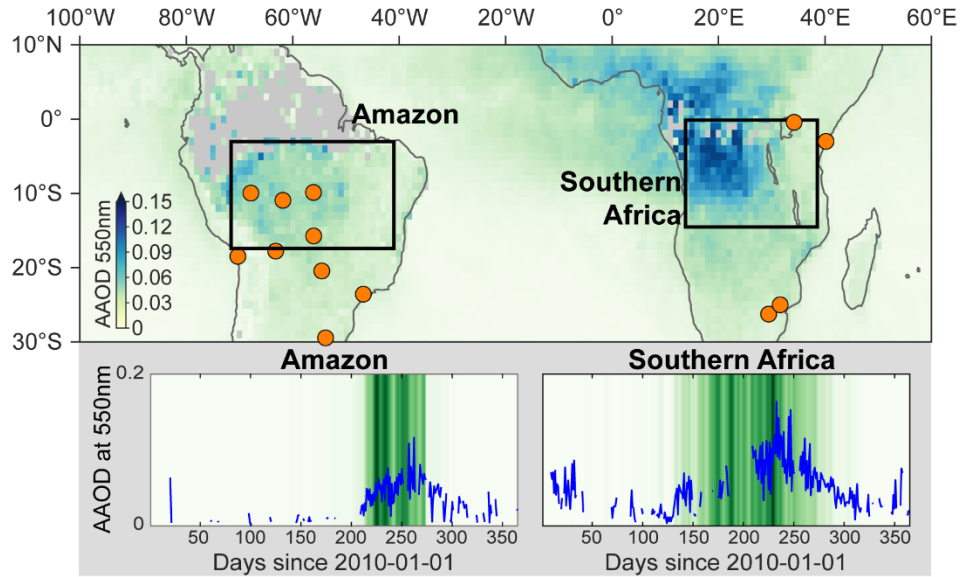
observations is found when a refractive index of  $0.3i$  is used. Based on the analysis, we correct the particle size and use the new refractive index in the two global models (ECHAM-HAM and SPRINTARS) to improve the modeled SSA (and MAC).

### **Supplementary Text 2. Predicting AAOD in African outflow for AeroCom models**

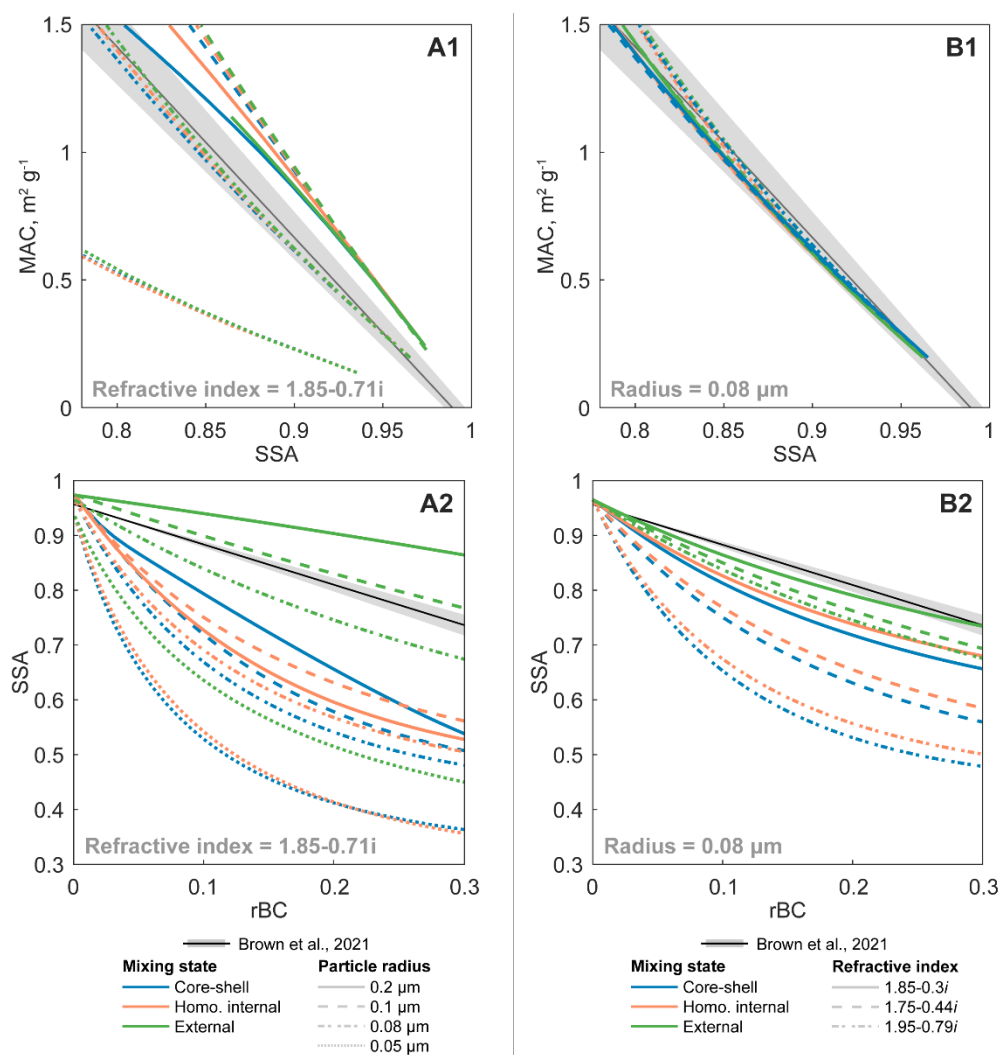
Following our previous study, we adopt a meta-model analysis to predict the African outflow AAOD for the AeroCom models (31). A linear regression is established for the outflow AAOD ( $AAOD_o$ ) as a function of emission ( $E_s$ ), lifetime ( $\tau_s$ ), and MAC ( $MAC_s$ ) in the source region using AeroCom model data (**Fig. S7A**). The general form of the regression can be written as:

$$AAOD_o = a \times E_s \tau_s MAC_s + b \times E_s \tau_s + c \times MAC_s + d$$

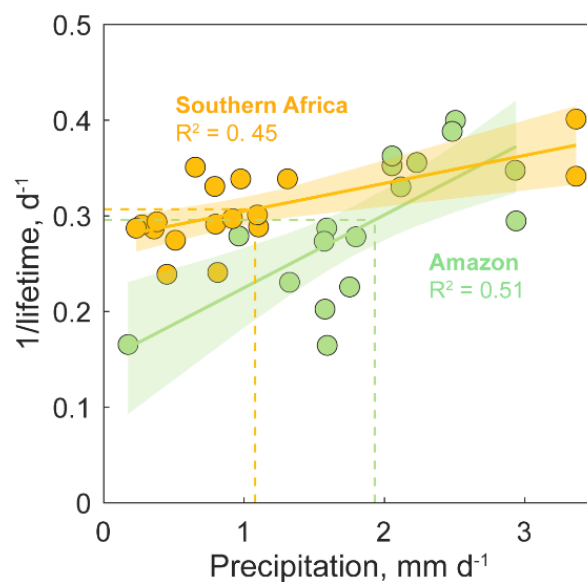
where  $a$ ,  $b$ ,  $c$ ,  $d$  are the coefficients obtained from the regression using all model data. The detailed formula derivation can be found in ref (31). Then, the constrained values for  $E_s$ ,  $\tau_s$ , and  $MAC_s$  are used to predict the outflow AAOD. In **Fig. S7B**, we validate the predicted AAOD and the original AeroCom model output against satellite observations and an improved agreement is found for the prediction based on our constrained results. This directly demonstrates the utility of our method to predict the AAOD outflow. In addition, it verifies the reliability of the three constrained components ( $E_s$ ,  $\tau_s$ , and  $MAC_s$ ) over the source region.



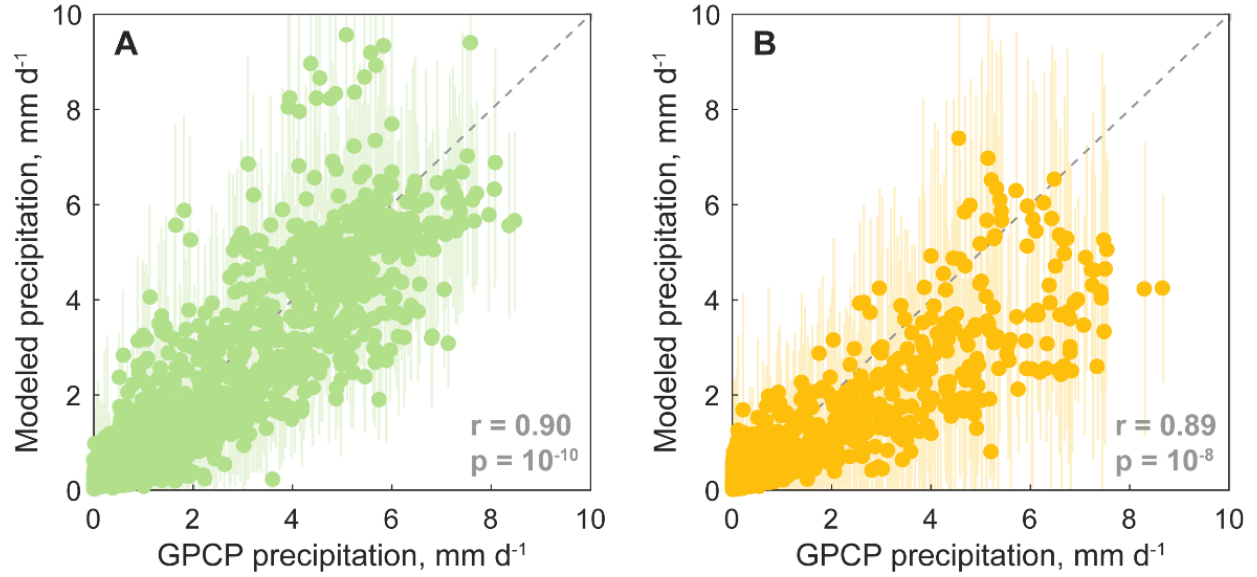
**Fig. S1 Geographical distribution of annual mean AAOD from POLDER-GRASP observations (top) and daily series of AAOD over the two focused fire regions (bottom).** The orange dots in the top map show the locations of AERONET monitoring sites considered for validating satellite datasets. The shaded scale in the background of the bottom diagrams reveals the normalized emission intensity based on GFED (with dark color indicating a high emission).



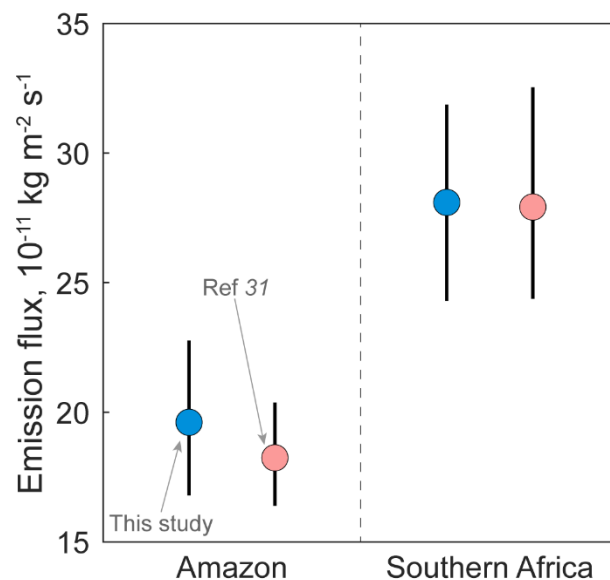
**Fig. S2 Relationships between MAC and SSA (A1, B1) and between SSA and rBC (A2, B2) affected by mixing state and particle size (A1, A2) and by mixing state and complex refractive index (B1, B2).** The rBC is calculated as  $\text{BC}:[\text{BC}+\text{OA}]$ . All the relationships are calculated at a wavelength of 550 nm based on the idealized Mie model for three mixing states: core-shell structure, homogeneous internal mixing, and external mixing (see Materials and Methods). Observations are taken from ref (24). Note that the observations by ref (24) consider background aerosols ( $< 15\%$ ), while our calculations are only for the mixture of BC and OA. In Fig. A1-A2, we assume a refractive index of  $1.85-0.71i$ , which is also used in the ECHAM-HAM model and represents moderate absorption in the model ensemble. In Fig. B1-B2, we assume a radius of  $0.08 \mu\text{m}$ , as suggested by ref (24). The three refractive indices in Fig. B1-B2 indicate the most absorbing ( $1.95-0.79i$ ) and least absorbing ( $1.75-0.44i$ ) in AeroCom models, and a suggested value based on in situ studies ( $1.85-0.30i$ ). There is an extreme case in Fig. A1 using the  $0.05 \mu\text{m}$  radius, which is typically smaller than the reported BBA size.



**Fig. S3 Relationships between modeled lifetime and modeled precipitation in the Amazon and Southern Africa.** Each dot represents the seasonally averaged data from a single model, with colors indicating the two fire regions. Lifetime is calculated as total burden divided by total emissions for BC and OA only. Solid lines are the regressions built between 1/lifetime and precipitation together with 95% confidence intervals (shaded area). Vertical dashed lines denote the GPCP observations for precipitation, and the horizontal dashed lines show the constrained 1/lifetime values. The R-squares ( $R^2$ ) of the regressions are shown for the two regions. Note that real lifetime regression uses both precipitation and the Angstrom Exponent (see Materials and Methods).

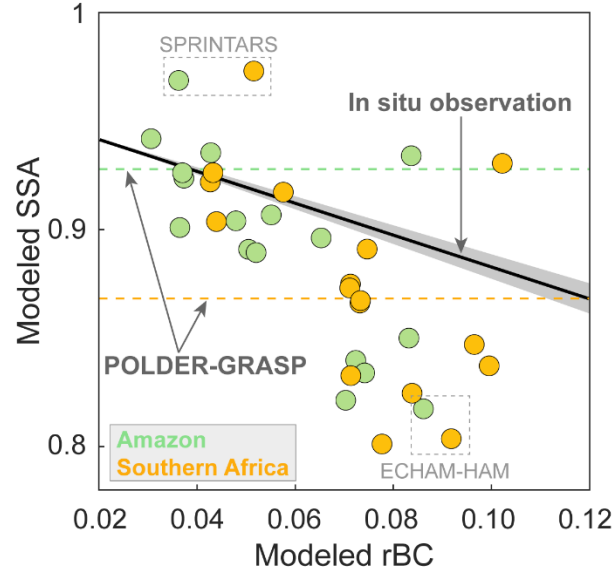


**Fig. S4 Comparisons of modeled precipitation with observations from GPCP for the Amazon (A) and Southern Africa (B).** Each dot represents the monthly average precipitation over a  $1^\circ \times 1^\circ$  grid box from either the GPCP dataset or the multi-model average from 17 AeroCom models. The vertical bars show the corresponding standard deviations of the 17 individual models. Dashed lines indicate the 1:1 ratio. The correlation coefficients ( $r$ ) and  $p$  value ( $p$ ) are shown for the two regions. Data are only considered for fire seasons.

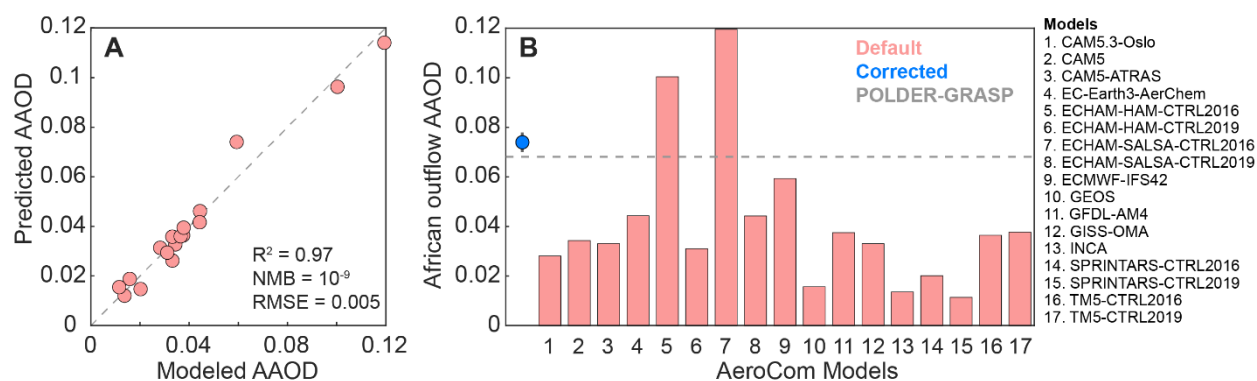


**Fig. S5 Comparisons of constrained emissions between this study and our previous work over the Amazon and Southern Africa.** The results from our previous work (31) are based on AOD and total aerosol extinction, which differs from this study. The error bars indicate the interquartile ranges considering all the uncertainty factors. Note that the emissions from ref (31) represent all aerosol species, while the results from this study are for BC+OA only (the latter is the dominant component of the former).

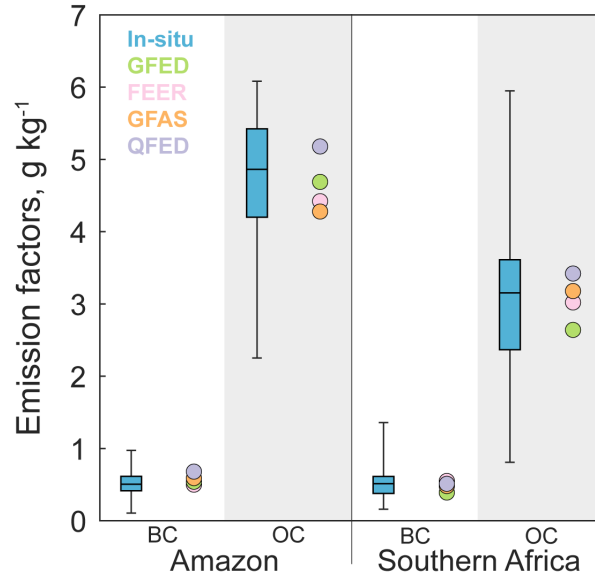




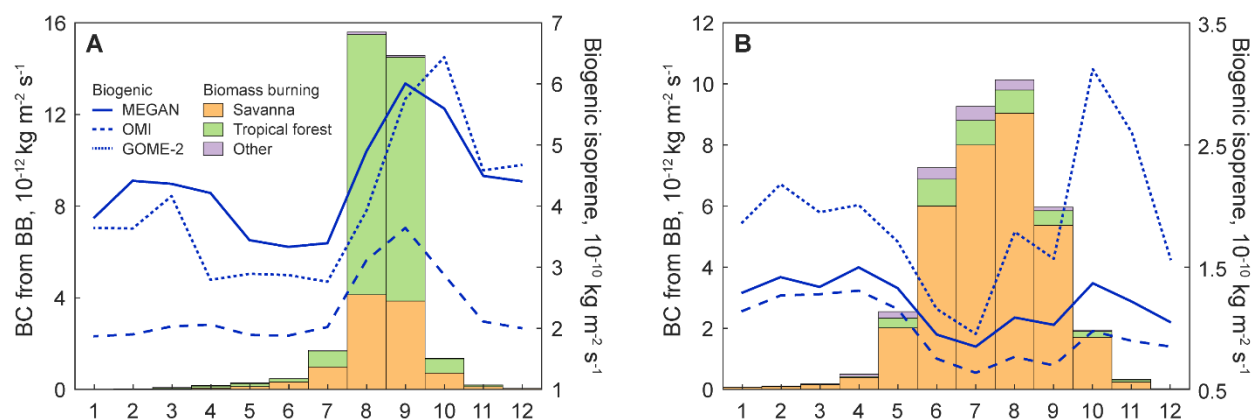
**Fig. S6 Comparison of modeled relationship between SSA and rBC with observed relationship.** The rBC is calculated as  $BC:[BC+OA]$ . Each data point represents the data averaged over the fire season, with the dot color indicating the two BB regions (green for Amazon and orange for Africa). The observed relationship from ref (24) is shown as a solid line with the 95% confidence interval (gray shaded area). SSA observations from POLDER-GRASP are shown as horizontal dashed lines. The ECHAM-HAM and SPRINTARS produce the most negative and positive SSA errors in the AeroCom model ensemble and are further corrected in this study.



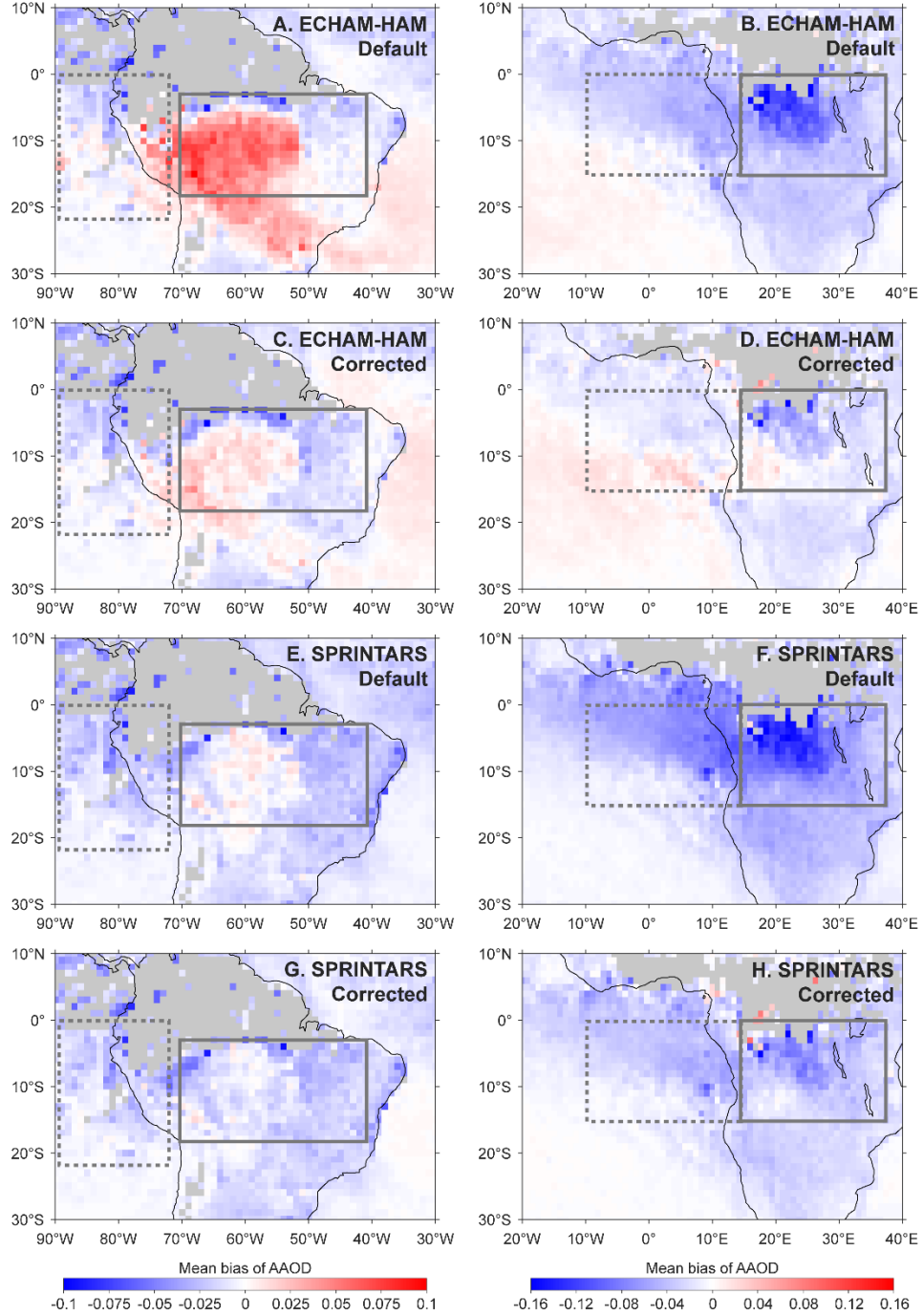
**Fig. S7 Linear regression for the African outflow AAOD based on AeroCom models (A) and the predicted outflow AAOD using the regression (B).** In Fig. A, the regression has the form of  $AAOD_o = A \times E_s \tau_s MAC_s + B \times E_s \tau_s + C \times MAC_s + D$ , where  $AAOD_o$  indicates the outflow AAOD;  $E_s$ ,  $\tau_s$ , and  $MAC_s$  are the total emission, lifetime, and MAC over the source region; and  $A$ ,  $B$ ,  $C$ , and  $D$  are the regression coefficients (see Supplementary Text 2). The metrics show the R-square ( $R^2$ ), normalized mean bias (NMB), and root mean square error (RMSE) for the regression. In Fig. B, constrained emission, lifetime, and MAC over the source region are adopted to the regression to predict the AAOD in the outflow area as a corrected case (blue dot), with the error bar showing the interquartile range of the prediction. The predicted AAOD value is compared with the default model data (red bars) and satellite observation (gray dashed line).



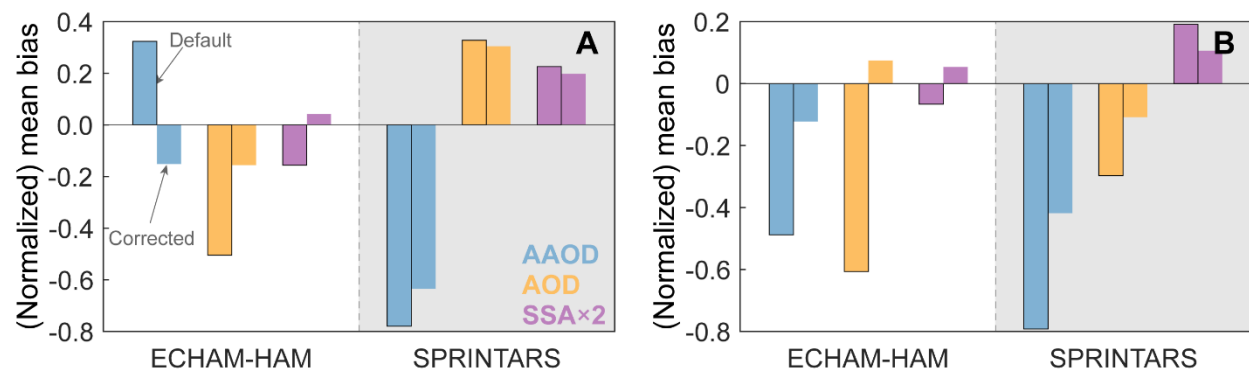
**Fig. S8 Comparisons of emission factors between emission inventories and in situ measurements over the Amazon and Southern Africa.** The in situ results are shown as box plots. The box plots show the 5-95% (whiskers), 25-75% percentile ranges (solid rectangles) and the median values (solid horizontal lines) based on the data collected from previous studies (Table S2). The data measured for tropical forest/deforestation fires and savanna/grassland fires are used in the analysis over the Amazon and Africa, respectively (Table S2).



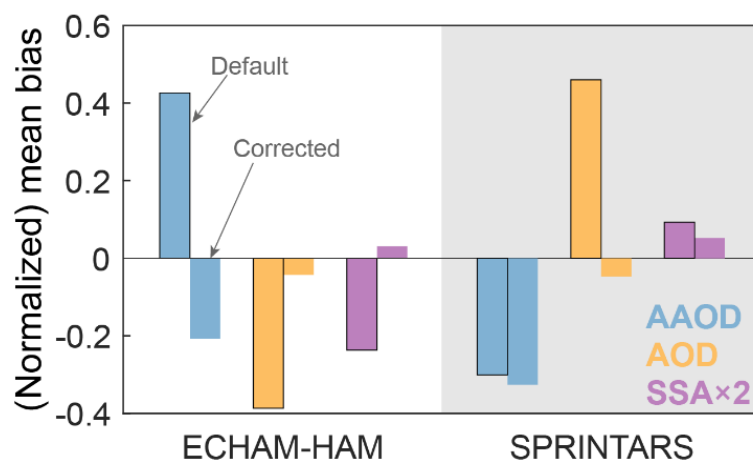
**Fig. S9 Monthly evolution of biomass burning BC emissions and biogenic isoprene emissions over the Amazon (A) and Southern Africa (B).** The emissions are shown as monthly flux averaged for each region. Biomass burning emissions are based on GFED4.1s (<http://www.globalfiredata.org/>), which are further divided into three fire types (savanna, tropical forest, and other). Biogenic emissions are obtained from one bottom-up estimate based on MEGAN model driven by ERA5 and two top-down estimates using constraints from OMI and GOME-2 satellite data (<https://emissions.aeronomie.be/>).



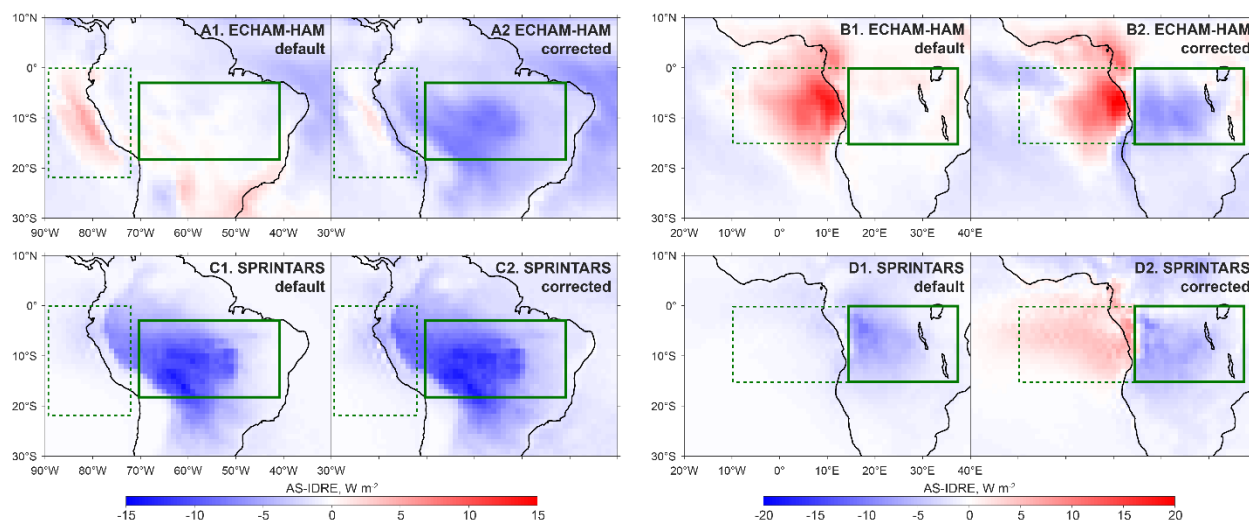
**Fig. S10 Seasonal mean bias for AAOD over the Amazon (A, C, E, G) and Africa (B, D, F, H) from ECHAM-HAM (A-D) and SPRINTARS (E-H).** The results for both default and corrected cases are shown. The data are collocated with POLDER-GRASP at  $1^\circ \times 1^\circ \times$  daily resolution. Gray areas indicate regions with no available observations. The gray boxes with solid edges show the source regions where we conduct the error analysis. The boxes with dashed edges show the corresponding outflow regions of focus in this study.



**Fig. S11 Seasonal mean bias of AAOD, AOD, and SSA for default and corrected simulations by ECHAM-HAM and SPRINTARS over the outflow regions of the Amazon (A) and Southern Africa (B).** The results are shown in the same format as Fig. 5 but for outflow regions as indicated in Fig. S10. Model data are collocated with the POLDER-GRASP dataset during the fire seasons.

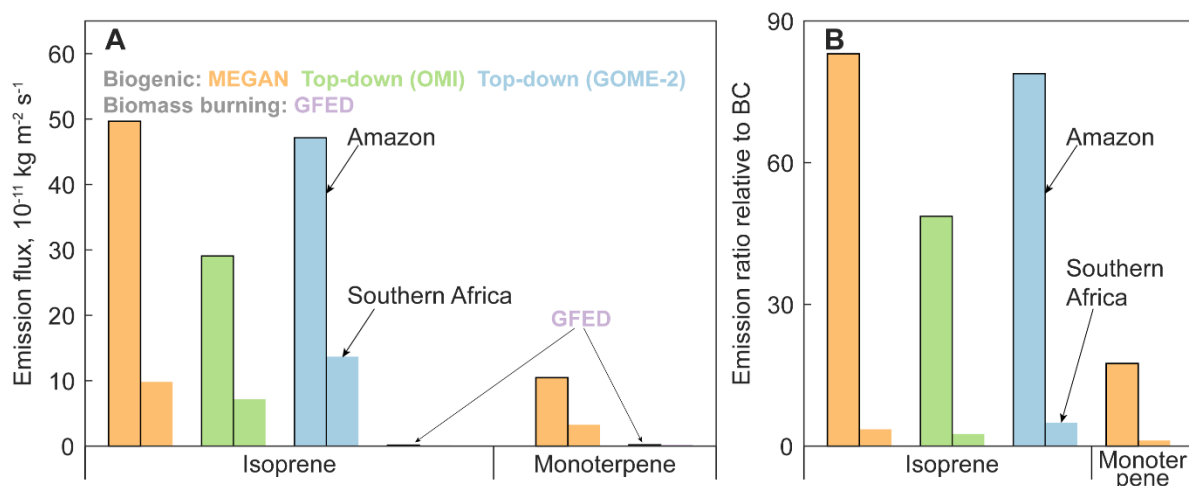


**Fig. S12 Seasonal mean bias of AAOD, AOD, and SSA for the default and corrected ECHAM-HAM and SPRINTARS simulations compared with AERONET data.** The results are shown in the same format as Fig. 5A, except for that the observations are taken from AERONET sites as shown in Fig. S1.

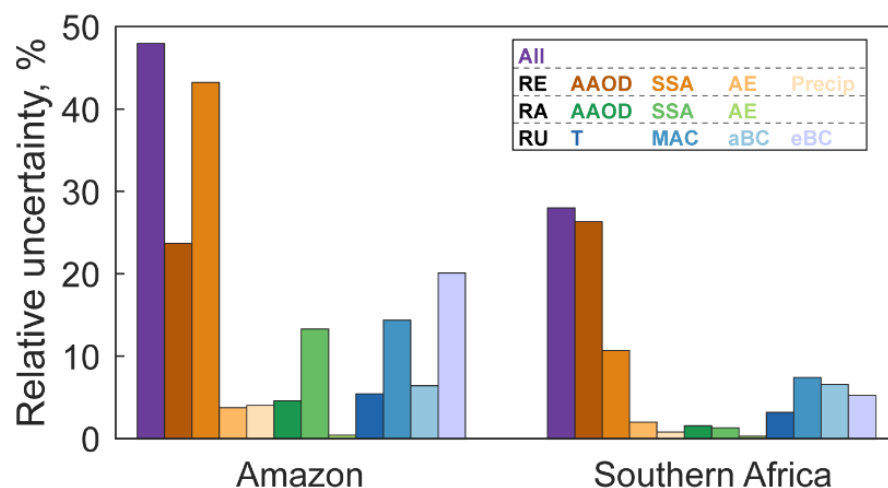


**Fig. S13 Seasonal mean IDRE over the Amazon (A, C) and Africa (B, D) from the ECHAM-HAM and SPRINTARS models for the default and corrected cases.** The all-sky IDREs (AS-IDRE) are shown as averages over the fire seasons. The green boxes show the source regions. The boxes with dashed edges indicate the corresponding outflow regions of focus in this study.

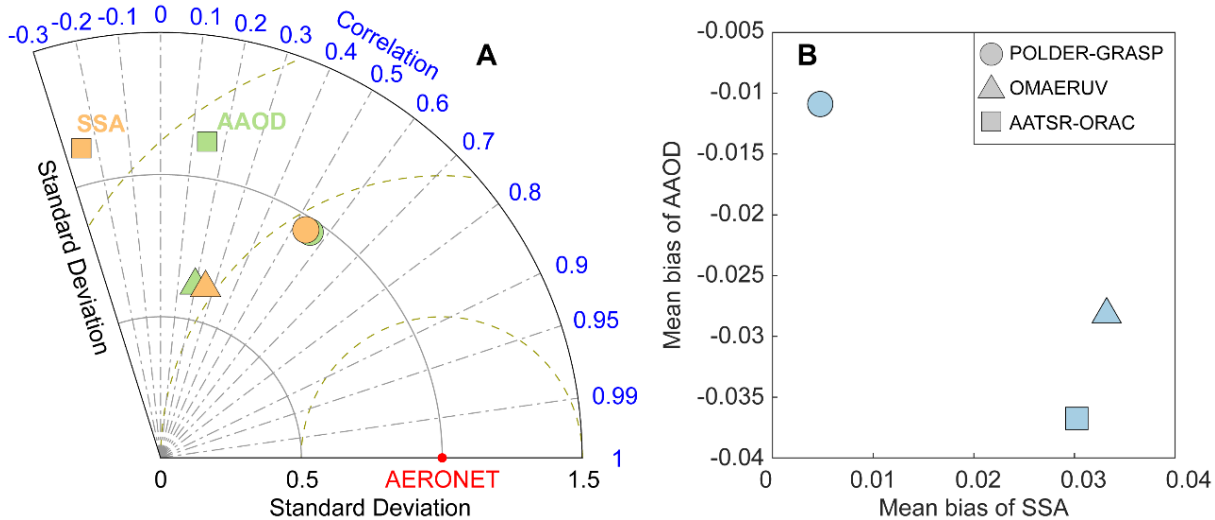




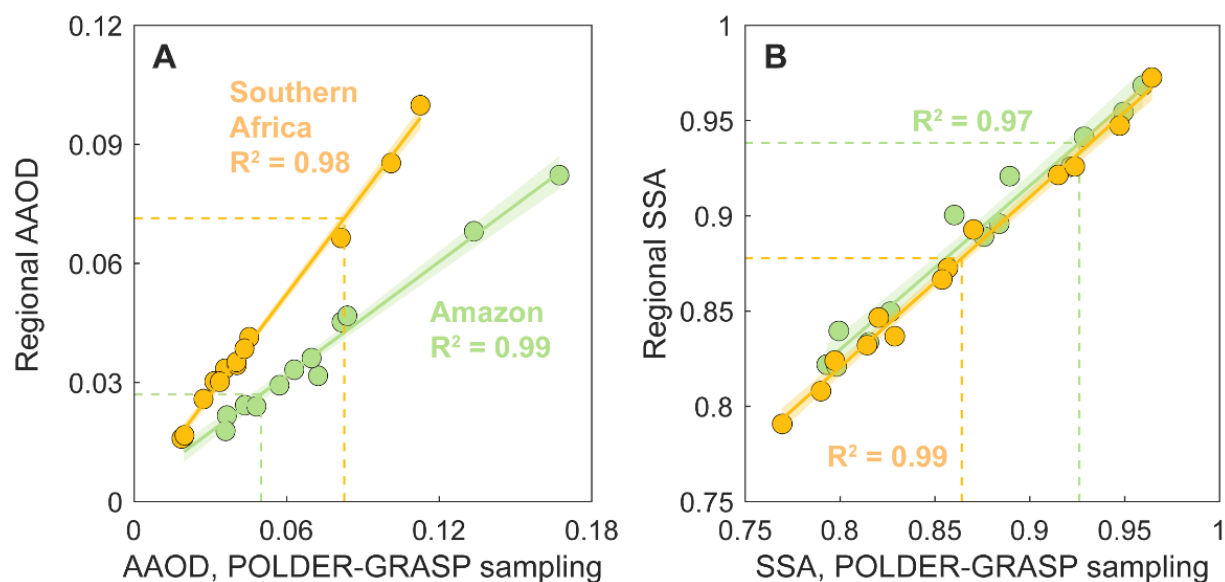
**Fig. S14 The emissions for precursor gas relevant to SOA formation.** A) The emissions of isoprene and monoterpene are shown for the Amazon (bar with a solid edge) and Southern Africa (bar without edge) for both biogenic and biomass sources. For biogenic emissions, one bottom-up (MEGAN) and two top-down (constrained by OMI and GOME-2 satellite observations, respectively) datasets are presented (72-74). For comparison, the GFED BB emissions are shown (which are much lower than biogenic emissions). All emissions are shown as the regional average flux over the fire seasons. B) The ratio of the biogenic isoprene and monoterpene emissions to constrained BC emissions. The sources of biogenic emissions are the same as Fig. A.



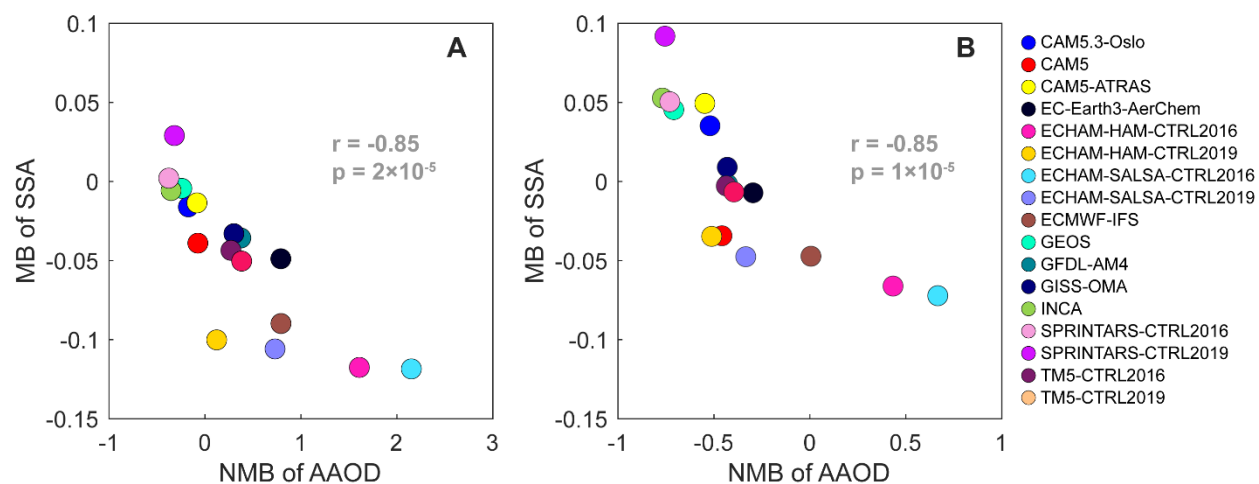
**Fig. S15 The relative uncertainties of constrained BC emissions due to all and individual uncertainty sources.** The uncertainties are shown as interquartile divided by median. The overall uncertainties are shown as purple bars (All). Individual uncertainty factors are considered for retrieval error (RE) of AAOD, SSA, AE, and precipitation; uncertainties of the regional averages (RA) for AAOD, SSA, and AE; the regression uncertainties (RU) for constrained lifetime (T), MAC, ambient BC:[BC+OA] (aBC), and emitted BC:[BC+OA] (eBC).



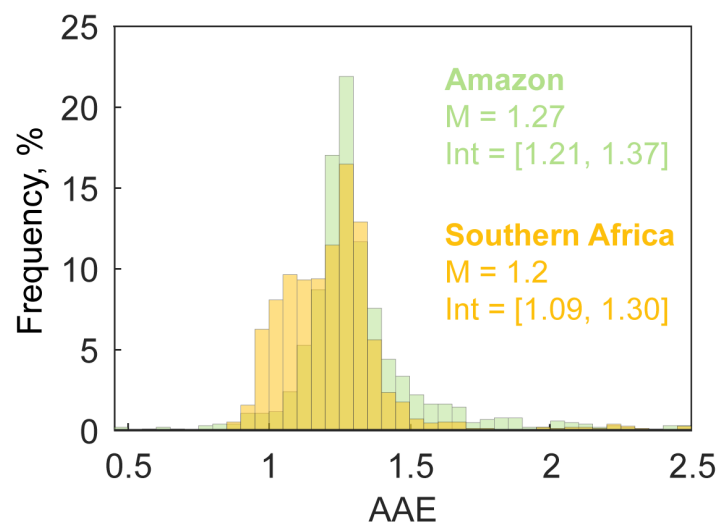
**Fig. S16 Validation of three satellite products against AERONET dataset.** The validation is shown as a Taylor diagram for AAOD (light green) and SSA (orange) in Fig. A and a scatter plot for mean bias in Fig. B. The shape of the symbols indicates different satellite products. All three satellite products and the AERONET dataset are collocated with each other during the fire season to ensure the same sampling.



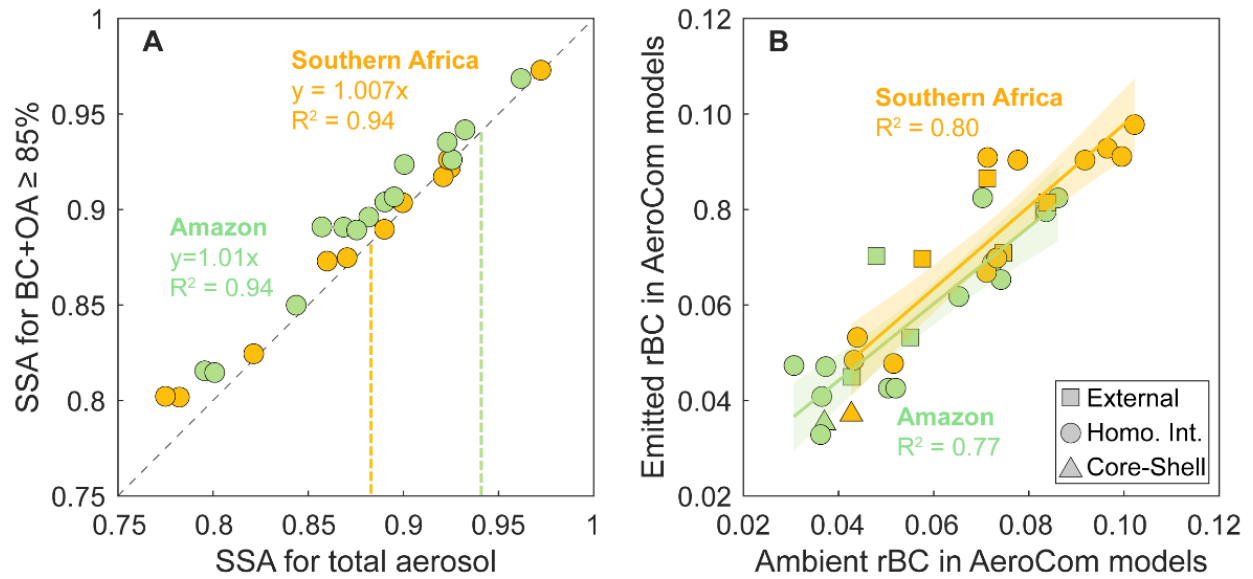
**Fig. S17 Relationship between averages of regional AAOD (A) and SSA (B) and POLDER-GRASP sampled values in AeroCom.** Each dot represents the average value from an individual model. For the sampled AAOD and SSA, model data during fire seasons are collocated with POLDER-GRASP on a daily basis. The solid lines show the linear regressions with 95% confidence intervals (shaded areas). The R-squares ( $R^2$ ) of the regressions are shown individually for the two fire regions. Vertical dashed lines show the average values of the raw POLDER-GRASP data, and the horizontal dashed lines indicate the predicted regional values.



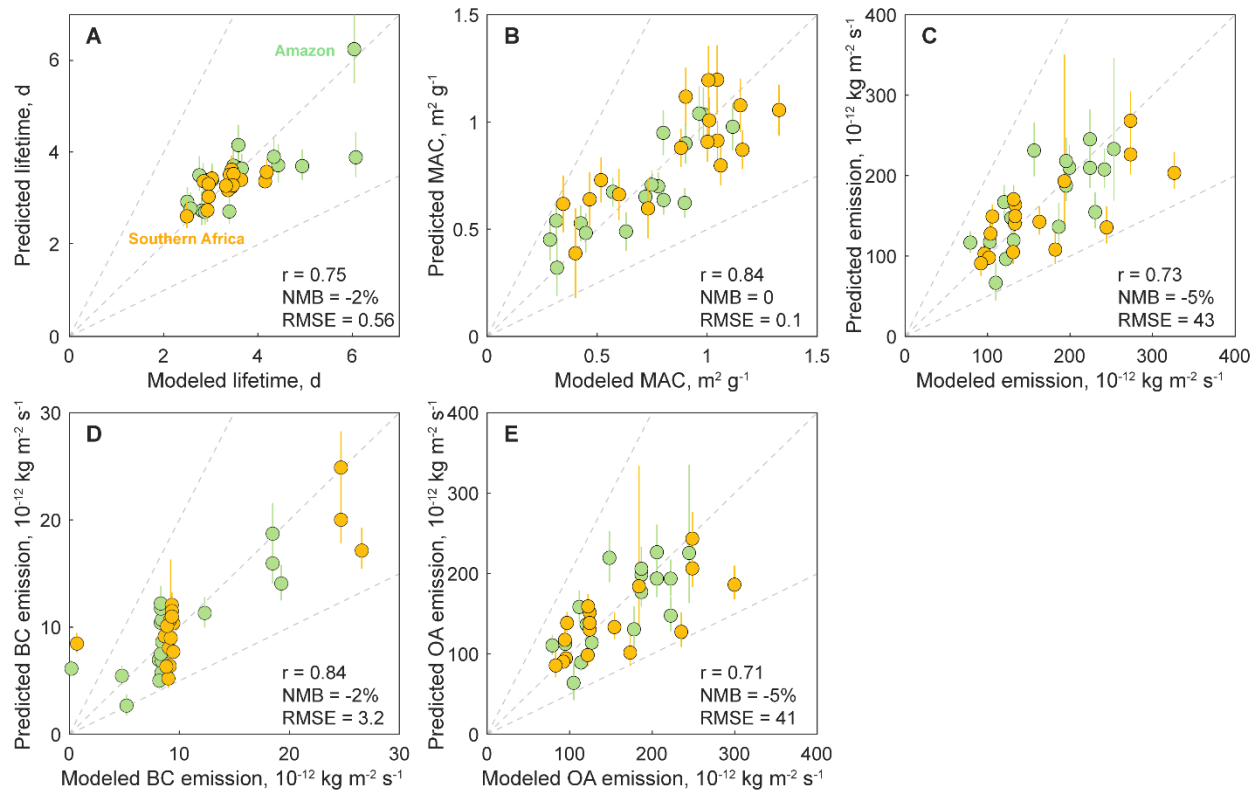
**Fig. S18 Modeled mean bias (MB) of SSA and normalized mean bias (NMB) of AAOD over the Amazon (A) and Southern Africa (B) during fire seasons.** Biases are calculated based on the non-located model data and reconstructed regional observations using POLDER-GRASP retrievals (see Fig. S17).



**Fig. S19 Distribution of the absorption angstrom exponent (AAE) from POLDER-GRASP over the Amazon and Southern Africa.** The AAE is calculated using the AAOD at 440/870 nm wavelength based on the daily data from individual grid cells during fire seasons. Statistics show the median (M) and interquartile ranges (Int) of the distributions.

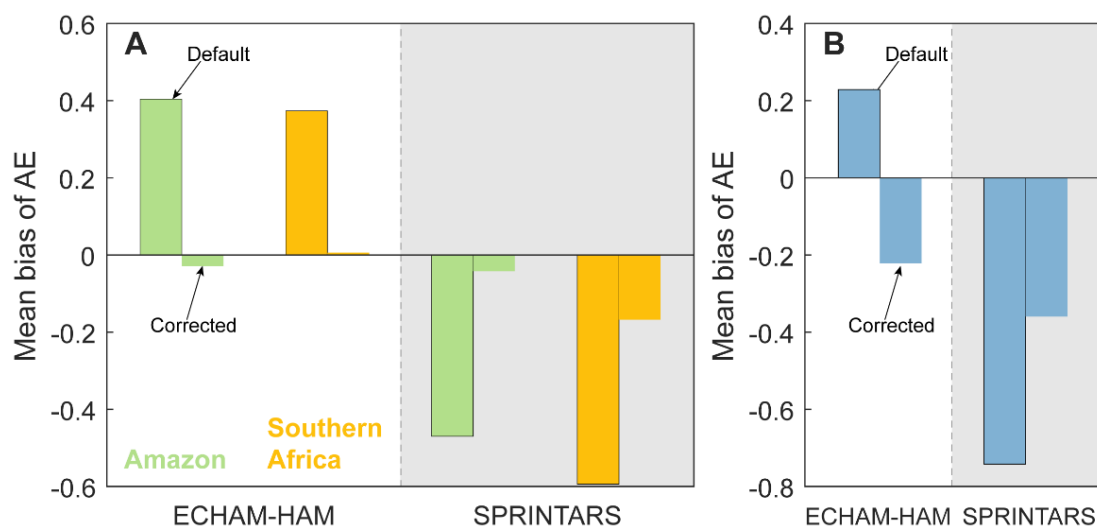


**Fig. S20 Relationships between SSA in all grid cells and SSA in grid cells with BC + OA  $\geq 85\%$  total aerosols (A) and the dependence of ambient rBC on emitted rBC (B).** The rBC is calculated as  $\text{BC}:[\text{BC}+\text{OA}]$ . The vertical dashed lines in Fig. A show the regional SSA observations from POLDER-GRASP. The black dashed line indicates the 1:1 line. In Fig. B, the linear regressions between emitted and ambient rBC are shown as solid lines with 95% confidence intervals (shaded areas).

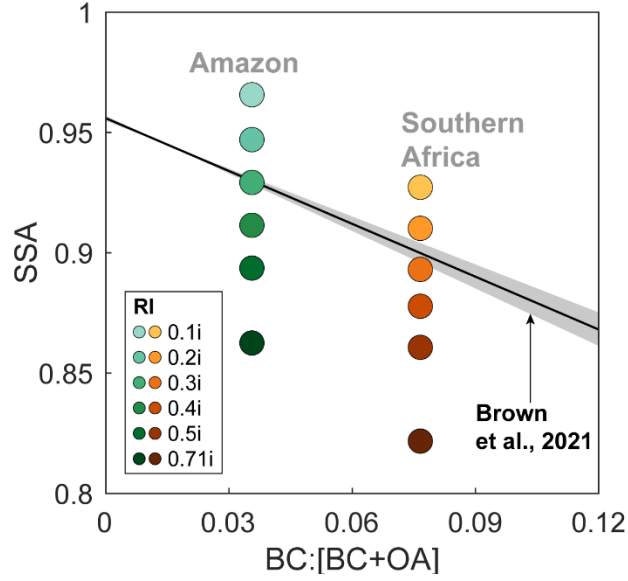


**Fig. S21 Comparisons of predicted lifetime (A), MAC (B), total emission (C), BC emission (D), and OA emission (E) using the constraining procedures with the original model data.** The predicted values for each model are estimated with all the other models following the constraining procedure. Vertical error bars denote the 95% prediction intervals. The dashed lines indicate the 1:1, 1:2, and 2:1 range. Metrics include the Pearson correlation coefficient ( $r$ ), normalized mean bias (NMB), and root mean square error (RMSE).

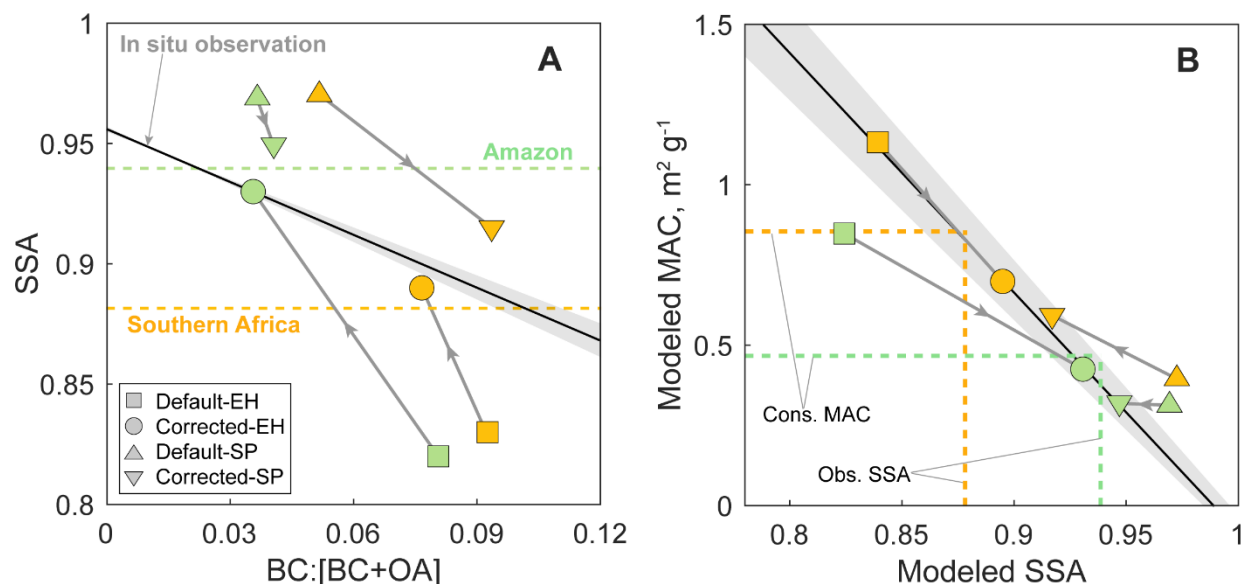




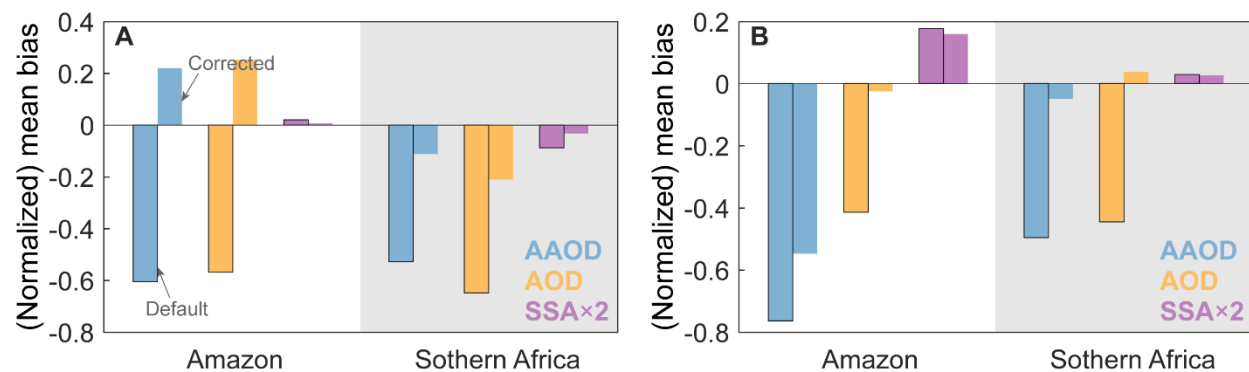
**Fig. S22 Seasonal mean bias for the Angstrom Exponent (AE) in ECHAM-HAM and SPRINTARS for default (bars with solid edges) and corrected simulations (bars without edges).** The model data are collocated and compared with POLDER-GRASP (A) and AERONET (B). Angstrom Exponent is calculated based on AOD at 440 and 550 nm wavelengths. The detailed configurations of the ambient particle size for the default and corrected cases can be found in Table S3. The comparison with POLDER-GRASP is conducted separately for the two regions as indicated by bar color. The AERONET observations are collected from the sites as shown in Fig. S1, which are mostly in the Amazon area.



**Fig. S23 Simulated changes in SSA in response to the different imaginary parts of the BC refractive index (RI) in the ECHAM-HAM model.** The colors of the circles denote data over two fire regions with color scales indicating different refractive indices. To highlight the impacts of the refractive index, the SSA is simulated using constrained emissions, modified particle size, and rescaled precipitation (Materials and Methods). In particular, the particle size is modified based on the modeled AE. The choice of refractive index of BC can affect the modeled AE, but the impact is generally small and would not fundamentally alter our results.



**Fig. S24 Modeled relationships between SSA and BC:[BC+OA] (A) and between MAC and SSA (B) using default and corrected configurations for ECHAM-HAM (EH) and SPRINTARS (SP).** The results are shown for the direct model output without collocations. For the corrected cases, models are corrected with rescaled particle size distribution, refractive index, modified precipitation, and constrained BC and OA emissions (see Materials and Methods). The black solid lines indicate the observations from ref (24) with 95% confidence intervals (gray shaded areas). Data for the two fire regions are shown in different colors. Regional observations of SSA based on POLDER-GRASP are shown as horizontal dashed lines in Fig. A and vertical dashed lines in Fig. B. The constrained MACs are shown as horizontal dashed lines in Fig. B. Note that the constrained MAC is predicted based on model relationship instead of the observed relationship as shown. The arrows in both plots indicate the direction from default to corrected simulations.



**Fig. S25 Seasonal mean bias of AAOD, AOD, and SSA from the default and corrected ECHAM-HAM simulations over the source (A) and outflow regions (B) of the Amazon and Southern Africa.** Model data are collocated and compared with POLDER-GRASP observation for 2009 fire season. The source and outflow regions are shown in Fig. S10.

**Table S1. Information on biomass burning aerosols in the 17 AeroCom models used in this study.** The experiment name (CTRL2016, CTRL2019) is added to the names of the models if they participate in both experiments. Note that the information is for biomass burning aerosols only.

| <b>Models</b>               | <b>Mixing state for BC</b>                                                                                                                 | <b>Emission</b> | <b>OA/OC</b> | <b>Refractive index</b>                              | <b>Ref</b> |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------|------------------------------------------------------|------------|
| <b>CAM5.3-Oslo</b>          | Internal mixing with Maxwell-Garnett rule                                                                                                  | CMIP5           | 2.6          | OA: 1.53-0.006 <i>i</i><br>BC: 1.95-0.79 <i>i</i>    | 75         |
| <b>CAM5</b>                 | Internal mixing                                                                                                                            | CMIP6           | 1.4          | OA: 1.53-0.005665 <i>i</i><br>BC: 1.95-0.79 <i>i</i> | 76         |
| <b>CAM5-ATRAS</b>           | Core-shell for internally-mixed BC particles                                                                                               | CMIP6           | 1.4          | OA: 1.53-0.005665 <i>i</i><br>BC: 1.95-0.79 <i>i</i> | 77         |
| <b>EC-Earth3-AerChem</b>    | Internal mixing with Maxwell-Garnett rule                                                                                                  | CMIP6           | 1.6          | OA: 1.53-0.00554 <i>i</i><br>BC: 1.85-0.71 <i>i</i>  | 78         |
| <b>ECHAM-HAM-CTRL2016</b>   | Internal mixing with volume-weighted average                                                                                               | 3.4×GFAS        | 1.4          | OA: 1.53-0.0055 <i>i</i><br>BC: 1.85-0.71 <i>i</i>   | 54         |
| <b>ECHAM-HAM-CTRL2019</b>   | Internal mixing with volume-weighted average                                                                                               | CMIP6           | 1.4          | OA: 1.53-0.0055 <i>i</i><br>BC: 1.85-0.71 <i>i</i>   | 54         |
| <b>ECHAM-SALSA-CTRL2016</b> | Internal mixing for each mode with volume-weighted average                                                                                 | 3.4×GFAS        | 1.4          | OA: 1.53-0.0055 <i>i</i><br>BC: 1.85-0.71 <i>i</i>   | 79         |
| <b>ECHAM-SALSA-CTRL2019</b> | Internal mixing for each mode with volume-weighted average                                                                                 | CMIP6           | 1.4          | OA: 1.53-0.0055 <i>i</i><br>BC: 1.85-0.71 <i>i</i>   | 79         |
| <b>ECMWF-IFS</b>            | External mixing                                                                                                                            | 3.4×GFAS        | 1.8          | OA: 1.5346-0.015 <i>i</i><br>BC: 1.75-0.45 <i>i</i>  | 80         |
| <b>GEOS</b>                 | External mixing                                                                                                                            | CMIP6           | 1.8          | OA: 1.53-0.005 <i>i</i><br>BC: 1.75-0.45 <i>i</i>    | 81         |
| <b>GFDL-AM4</b>             | All aerosols are externally mixed except for sulfate and black carbon which are internally mixed by volume weighting of refractive indices | CMIP6           | 1.4          | OA: 1.53-0.006 <i>i</i><br>BC: 1.75-0.44 <i>i</i>    | 82         |
| <b>GISS-OMA</b>             | External mixing, with a BC absorption amplification factor of 1.5                                                                          | CMIP6           | 1.4          | OA: 1.527-0.014 <i>i</i><br>BC: 1.85-0.71 <i>i</i>   | 83         |
| <b>INCA</b>                 | Internal mixing for BC with Maxwell-Garnett to compute its refractive index                                                                | CMIP6           | 1.4          | OA: 1.53-0.0055 <i>i</i><br>BC: 1.75-0.44 <i>i</i>   | 84,85      |
| <b>SPRINTARS-CTRL2016</b>   | Internal mixing of BC and OA with volume-weighted average for effective refractive index                                                   | GFED3.1         | 2.6          | OA: 1.53-0.006 <i>i</i><br>BC: 1.75-0.44 <i>i</i>    | 55         |

|                           |                                                                                          |       |     |                                                     |       |
|---------------------------|------------------------------------------------------------------------------------------|-------|-----|-----------------------------------------------------|-------|
| <b>SPRINTARS-CTRL2019</b> | Internal mixing of BC and OA with volume-weighted average for effective refractive index | CMIP6 | 2.6 | OA: 1.53-0.006 <i>i</i><br>BC: 1.75-0.44 <i>i</i>   | 55    |
| <b>TM5-CTRL2016</b>       | Internal mixing with Maxwell-Garnett rule                                                | CMIP6 | 1.6 | OA: 1.53-0.00554 <i>i</i><br>BC: 1.85-0.71 <i>i</i> | 78,86 |
| <b>TM5-CTRL2019</b>       | Internal mixing with Maxwell-Garnett rule                                                | CMIP6 | 1.6 | OA: 1.53-0.00554 <i>i</i><br>BC: 1.85-0.71 <i>i</i> | 78,86 |

**Table S2. Emission factors of BC, OC, and OC/BC ratio for biomass burning in previous studies.** Data (unit: g/kg dry matter burned) are collected from in situ measurements or emission inventories for tropical forest/deforestation and savanna/grassland fires regarding the major biomes burned in the Amazon and Southern Africa, respectively. Values used in the four emission inventories are shown in italics for comparison. The mean values and standard errors (SE) for emission factors of BC and OC across in-situ studies are used to estimate the OC/BC ratio and its uncertainty (see Materials and Methods).

| Tropical forest/deforestation |             |             |             | Savanna/grassland |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------------|-------------|-------------|-------------|
| Ref                           | BC          | OC          | OC/BC       | Ref               | BC          | OC          | OC/BC       |
| <i>87</i>                     | 0.99        | 3.56        | 3.6         | <i>98</i>         | 0.69        | 3.16        | 4.56        |
| <i>88</i>                     | 0.96        | 6.08        | 6.34        | <i>88</i>         | 0.16        | 2.82        | 17.31       |
| <i>89</i>                     | 0.59        | 6.08        | 10.38       | <i>99</i>         | 0.37        | 3.28        | 8.88        |
| <i>90</i>                     | 0.47        |             |             | <i>99</i>         | 0.45        |             |             |
| <i>91</i>                     | 0.66        | 5.2         | 7.88        | <i>100</i>        | 0.26        |             |             |
| <i>92</i>                     | 0.46        |             |             | <i>101</i>        | 0.61        |             |             |
| <i>93</i>                     | 0.19        | 1.53        | 7.96        | <i>102</i>        | 0.59        | 1.61        | 2.73        |
| <i>94</i>                     | 0.52        | 4.71        | 9.06        | <i>89</i>         | 0.52        | 3.53        | 6.82        |
| <i>95</i>                     | 0.02        | 5           | 263.16      | <i>90</i>         | 0.78        |             |             |
| <i>96</i>                     | 0.38        |             |             | <i>103</i>        | 0.59        | 1.61        | 2.73        |
| <i>97</i>                     | 0.51        | 4.4         | 8.63        | <i>104</i>        | 0.39        | 2.3         | 5.9         |
| <b>Mean</b>                   | <b>0.52</b> | <b>4.57</b> | <b>8.76</b> | <i>105</i>        | 0.24        | 0.7         | 2.92        |
| <b>SE</b>                     | <b>0.09</b> | <b>0.53</b> |             | <i>105</i>        | 0.47        | 3           | 6.38        |
| <i>GFED</i>                   | <i>0.52</i> | <i>4.71</i> | <i>9.06</i> | <i>106</i>        | 0.5         | 2.84        | 5.69        |
| <i>FEER</i>                   | <i>0.51</i> | <i>4.4</i>  | <i>8.63</i> | <i>94</i>         | 0.61        | 5.79        | 9.49        |
| <i>GFAS</i>                   | <i>0.57</i> | <i>4.3</i>  | <i>7.54</i> | <i>107</i>        | 1.29        | 3.68        | 2.85        |
| <i>QFED</i>                   | <i>0.66</i> | <i>5.2</i>  | <i>7.88</i> | <i>108</i>        | 1.43        | 2.44        | 1.7         |
|                               |             |             |             | <i>109</i>        | 0.16        |             |             |
|                               |             |             |             | <i>110</i>        | 0.53        |             |             |
|                               |             |             |             | <i>111</i>        |             | 3.56        |             |
|                               |             |             |             | <i>95</i>         | 0.13        | 0.82        | 6.31        |
|                               |             |             |             | <i>97</i>         | 0.37        | 3.28        | 8.88        |
|                               |             |             |             | <b>Mean</b>       | <b>0.51</b> | <b>2.66</b> | <b>5.19</b> |
|                               |             |             |             | <b>SE</b>         | <b>0.07</b> | <b>0.31</b> |             |
|                               |             |             |             | <i>GFED</i>       | <i>0.37</i> | <i>2.62</i> | <i>7.08</i> |
|                               |             |             |             | <i>FEER</i>       | <i>0.53</i> | <i>3</i>    | <i>5.66</i> |
|                               |             |             |             | <i>GFAS</i>       | <i>0.46</i> | <i>3.2</i>  | <i>6.96</i> |
|                               |             |             |             | <i>QFED</i>       | <i>0.48</i> | <i>3.4</i>  | <i>7.08</i> |

**Table S3. Model configurations for ECHAM-HAM and SPRINTARS.**

|                  | Models    | Emission                                                                                                                                                                                         | Precipitation                        | RI <sup>a</sup>    | Particle size <sup>c</sup>                                                                                                                                       |
|------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Default</b>   | ECHAM-HAM | CMIP6 for BB and anthropogenic; AeroCom-II for biogenic emissions; Prescribed SOA productions; Online calculations for dust and sea salt                                                         | See ref (65)                         | 1.85-0.71 <i>i</i> | BB particles are emitted at 75 nm, with microphysical processes calculated by the modal scheme M7. Hygroscopic growth is based on the Kappa- Köhler theory (65). |
|                  | SPRINTARS | GFED4.1s for BB; CMIP6 for anthropogenic emission; GEIA for biogenic emissions; Online SOA production based on the oxidation of isoprene and terpenes; Online calculations for dust and sea salt | See ref (55)                         | 1.75-0.44 <i>i</i> | BB particles are emitted at 100 nm without growth of dry size. Hygroscopic growth refers to ref (55).                                                            |
| <b>Corrected</b> | ECHAM-HAM | BC and OA (both primary and secondary) emissions are corrected to our constrained results                                                                                                        | Scaled by 3.3 in Africa <sup>b</sup> | 1.85-0.3 <i>i</i>  | Increase the emitted size to 200 nm and scale the ambient size by 1.1                                                                                            |
|                  | SPRINTARS | BC and OA (both primary and secondary) emissions are corrected to our constrained results                                                                                                        | Scaled by 0.3 <sup>b</sup>           | 1.75-0.3 <i>i</i>  | 100 nm emitted size with no hygroscopic growth                                                                                                                   |

**a.** RI (refractive index) is shown for BC, and the RI for OA uses the model default values (see Table S1). The real parts of the BC refractive index in the corrected cases are the same as the default values as they have small impacts on the results.

**b.** The scaling factors based on modeled precipitation errors are applied to the wet removal directly.

**c.** The particle size refers to the number median radius.



**Table S4. Comparison of the all-sky instantaneous direct radiative effect (unit:  $\text{W m}^{-2}$ ) for the default and corrected simulations in the two global models.** Data are presented as regional and seasonal average. The source and outflow regions are indicated in Fig. S10.

| <b>Region</b> | <b>Model</b> | <b>Source region</b> |           | <b>Outflow region</b> |           |
|---------------|--------------|----------------------|-----------|-----------------------|-----------|
|               |              | Default              | Corrected | Default               | Corrected |
| Amazon        | ECHAM-HAM    | -0.34                | -4.35     | 0.49                  | -1.36     |
|               | SPRINTARS    | -5.72                | -6.70     | -1.23                 | -1.80     |
| Africa        | ECHAM-HAM    | 0.04                 | -3.34     | 5.37                  | 3.15      |
|               | SPRINTARS    | -3.54                | -3.56     | -1.25                 | 2.44      |

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